

Pranav Deshmukh

Embedded Systems & IoT Firmware Engineer • 4 Years Experience • MERIQ Index: 69% (Developing)

■ pranav.deshmukh720@example.com | ■ +1 (427) 562-4694 | ■ Boston, MA

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, ARM Cortex, PCB Debugging, Python, MQTT, FreeRTOS, Embedded Linux

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 49%.
- Led a squad of 7 engineers delivering scalable microservices processing 36M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 37%.

Associate Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2022 - 2023)

- Designed and maintained ARM Cortex data endpoints with 99% SLA uptime.
- Optimized database indexing and caching layers using PCB Debugging saving \$20k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [C/C++, ARM Cortex, PCB Debugging, Python]

Engineered end-to-end production solution leveraging C/C++ and ARM Cortex with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [PCB Debugging, Python, MQTT, FreeRTOS]

Architected high-resilience system integrating PCB Debugging with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Cybersecurity and Information Assurance

ETH Zurich — 2018 - 2022 (Score: 3.45 / 4.0)